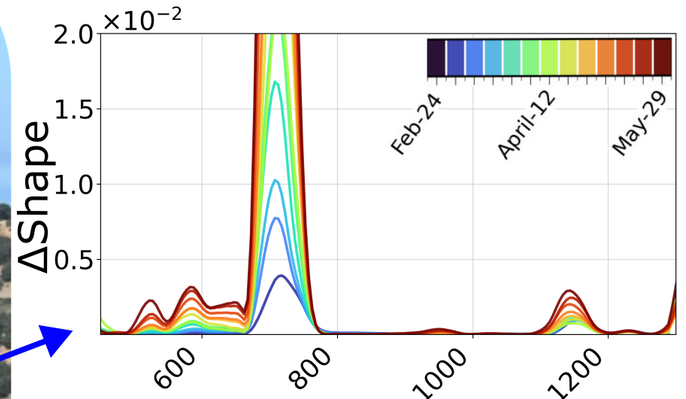
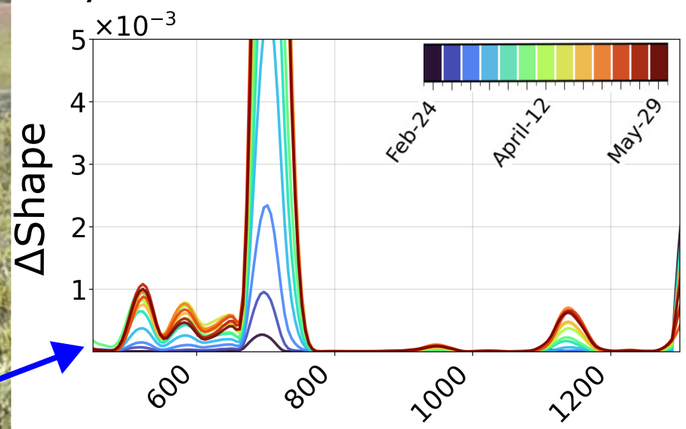
Identifying *where* and *when* biological information emerges in imaging spectroscopy data

- Independent shape-based operators converge on class-discriminative spectral segments.
- Collinearity of adjacent bands was leveraged to locate informative spectral segments.
- Shape divergence among vegetation types was strongest in the red-edge spectral region.
- Vegetation separability varied seasonally indicating importance of phenology.
- Classification remained robust with weekly to monthly could-free acquisitions.

Oak woodland vs. all 3 PFTs



Coyote brush scrub vs. PFT mean



Non-native grassland vs. PFT mean

